

Linear functions

A linear function has a constant rate of change.

$$f(x) = mx + b$$

What do m and b tell us about the graph?

Recognizing a constant rate

A function takes these values at equally spaced inputs.

$$f(0) = 7, \quad f(1) = 4, \quad f(2) = 1$$

Are these values consistent with a linear function?

Unequally spaced inputs

The points belong to one function.

$(1, 4), (3, 10), (6, 19)$

Are the rates between consecutive points equal?

Slope

For two distinct inputs, slope is the change in output divided by the change in input.

$$m = \frac{y_2 - y_1}{x_2 - x_1}$$

Why must the input difference be nonzero?

Calculating slope

A line passes through these points.

$$(-2, 7), \quad (4, -5)$$

What is its slope?

Slope units

A tank contains 80 liters at minute 2 and 56 liters at minute 8. Its volume changes linearly.

What is the rate of change, including units?

Increasing or decreasing

$$f(x) = -\frac{3}{2}x + 6$$

Is this function increasing, decreasing, or constant?

A constant function

$$f(x) = 9$$

How does its output change when the input increases?

A vertical line

$$x = 4$$

Why is this line not the graph of a function of x ?

The initial value

A machine costs \$240 to rent plus \$18 per hour.

$$C(h) = 240 + 18h$$

What does the vertical intercept represent?

Point-slope form

A line has slope 3 and passes through (2, -4).

$$y - y_1 = m(x - x_1)$$

What equation describes the line?

From two points

A linear function satisfies these conditions.

$$f(-1) = 5, \quad f(3) = -3$$

What is its formula in slope-intercept form?

From standard form

$$3x + 2y = 12$$

What are the slope and vertical intercept?

Graphing a line

$$f(x) = \frac{2}{3}x - 4$$

How can the intercept and slope locate points for a sketch?

Finding intercepts

$$f(x) = -4x + 12$$

Where does the graph meet each coordinate axis?

Parallel lines

Distinct nonvertical parallel lines have equal slopes.

$$y = 2x - 5$$

What equation describes the parallel line through (3, 4)?

Perpendicular lines

Two nonvertical perpendicular lines have slopes whose product is -1.

$$y = -\frac{2}{3}x + 1$$

What is the slope of a perpendicular line?

A perpendicular equation

$$y = 4x + 7$$

What perpendicular line passes through $(-2, 3)$?

Horizontal and vertical lines

One line is horizontal and passes through $(2, -3)$.

What perpendicular line passes through the same point?

An intersection

$$f(x) = 2x + 1, \quad g(x) = -x + 7$$

At what point do these graphs intersect?

Same slope, different intercepts

$$f(x) = 3x - 2, \quad g(x) = 3x + 5$$

Can their graphs intersect?

Linear change versus a ratio

$$f(x) = 4x, \quad x > 0$$

Is the ratio $f(x + 1)/f(x)$ constant?

A model domain

A taxi fare is \$5 plus \$2 per mile. Let d be distance traveled.

What domain is reasonable for the fare function?